
Session 4 | Linear Algebra II

Exercise sheet

Work through these in order: they start with finding eigenvalues and build toward PCA and the stability of dynamical systems. The recurring idea is that eigenvalues and eigenvectors reveal the “natural directions” of a matrix, and that is what powers dimensionality reduction and tells us how neural systems evolve in time.

Exercise 1 Let $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

- (a) Verify that $(1, 1)$ is an eigenvector, and give its eigenvalue.
 - (b) In one sentence, what does it mean for a vector to be an eigenvector of a matrix?
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Exercise 2 Find both eigenvalues of $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ by solving the characteristic equation $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$.

Exercise 3 For the same matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, find an eigenvector corresponding to the eigenvalue $\lambda = 2$.

Exercise 4 Still for $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$:

- (a) Compute the trace and the determinant.
 - (b) Verify that they equal $\lambda_1 + \lambda_2$ and $\lambda_1 \lambda_2$ respectively, using the eigenvalues you found.
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Exercise 5 Let $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ (symmetric).

- (a) Find its eigenvalues.
 - (b) Find an eigenvector for each, and check that the two eigenvectors are orthogonal.
 - (c) Why is orthogonality guaranteed for a symmetric matrix?
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Exercise 6 A discrete linear system repeatedly applies a matrix to its state.

The matrix has eigenvalues 0.9 (along direction $\mathbf{u}$) and 0.5 (along $\mathbf{v}$). The initial state has coefficient 1 along each eigenvector.

- (a) What are the two coefficients after 1 step?
- (b) After 10 steps? (You may leave them as powers or evaluate.)

- (c) Which mode dominates the long-run behaviour, and why?
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Exercise 7 *Principal Component Analysis of two correlated variables.*

The covariance matrix is $\mathbf{C} = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$.

- (a) Find its eigenvalues.
 - (b) Give the direction of the first principal component.
 - (c) What fraction of the total variance does the first component capture?
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Exercise 8 A PCA of neural data returns the eigenvalues 12, 6, 1.5, 0.5 (largest first).

- (a) Compute the total variance and the fraction explained by the first component.
 - (b) How many components are needed to capture at least 90% of the variance?
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Exercise 9 A covariance matrix has eigenvalues 8, 5, 2 (you are not told the matrix itself).

- (a) What is the total variance, and which single number of the matrix equals it?
 - (b) Can any eigenvalue of a covariance matrix be negative? Explain.
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Exercise 10 For each symmetric matrix, decide whether it *could* be a covariance matrix, based only on its eigenvalues, and say why:

- (a) eigenvalues 3 and 1;
 - (b) eigenvalues 4 and -2 ;
 - (c) eigenvalues 2 and 0.
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Exercise 11 *A recurrent network updates its activity as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t$.*

The connectivity matrix $\mathbf{A}$ has eigenvalues 0.95 and 1.05.

- (a) Is the network stable (does activity settle to zero)?
 - (b) Describe what happens to the activity over many steps.
 - (c) If both eigenvalues were 0.95 instead, would it be stable?
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Exercise 12 A continuous neural population obeys $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, and $\mathbf{A}$ has the complex eigenvalues $-1 \pm 2i$.

- (a) Is the system stable? Which part of the eigenvalue tells you?
 - (b) What behaviour does the imaginary part produce?
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Exercise 13 A continuous system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ has eigenvalues 2 (along $\mathbf{u}$) and -3 (along $\mathbf{v}$). After a long time, which direction does the state align with, and does it grow or decay?

Exercise 14 *PCA on a large population recording.*

Recording 100 neurons, PCA shows that the first 3 components explain 85% of the variance.

- (a) What does this say about the dimensionality of the population activity?
 - (b) Why is this useful for analysis and visualisation?
 - (c) Give one reason why choosing components by variance can be misleading.
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Exercise 15 A symmetric matrix can be written as $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^\top$ with $\mathbf{D}$ diagonal.

- (a) What do the columns of $\mathbf{P}$ represent?
- (b) What are the diagonal entries of $\mathbf{D}$?
- (c) Geometrically, what does applying $\mathbf{A}$ then do to space?